

JKU Visual Data Science Lab

AI and Visualization Lab WS26

AI4VIS Practical Examples

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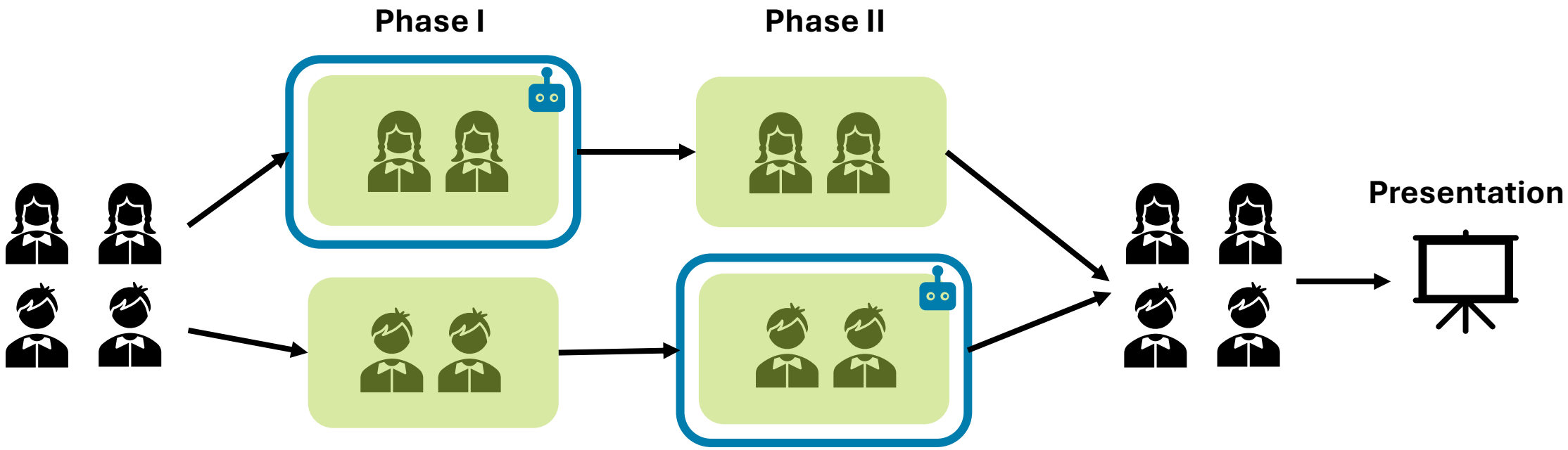
VIS4AI

AI4VIS

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Reminder

Explore how Generative AI can support and influence the visualization design and analysis process.



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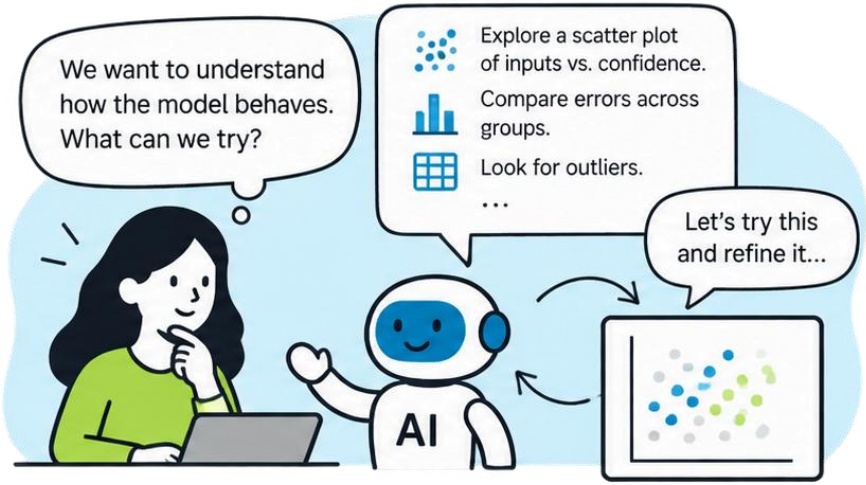
GenAI as Tool vs. GenAI as Collaborator

GenAI as Tool



Humam asks, GenAI implements

GenAI as Collaborator



Human and GenAI work together

The right approach depends on the context you're incorporating GenAI in.

In either case, the assignment requires reflection on GenAI contribution.

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#1 GenAI as Tool to Transform Your Data

GenAI can change what we are able to analyze.

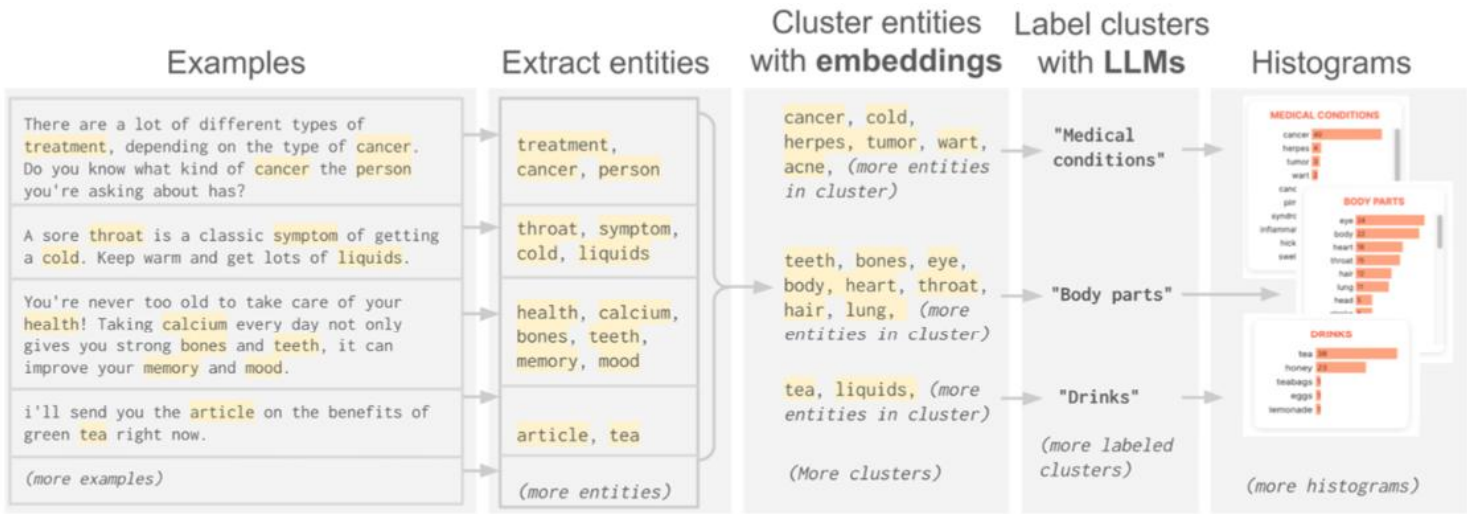
How to use?

Reminder: if we have high-dimensional unstructured data, GenAI can help structuring it.

One more example:

Interactive histograms to explore LLM answers to medical questions.

Responses were given a structure using semantic labels by an LLM.



Automatic Histograms: Leveraging Language Models for Text Dataset Exploration
Reif et al. 2024

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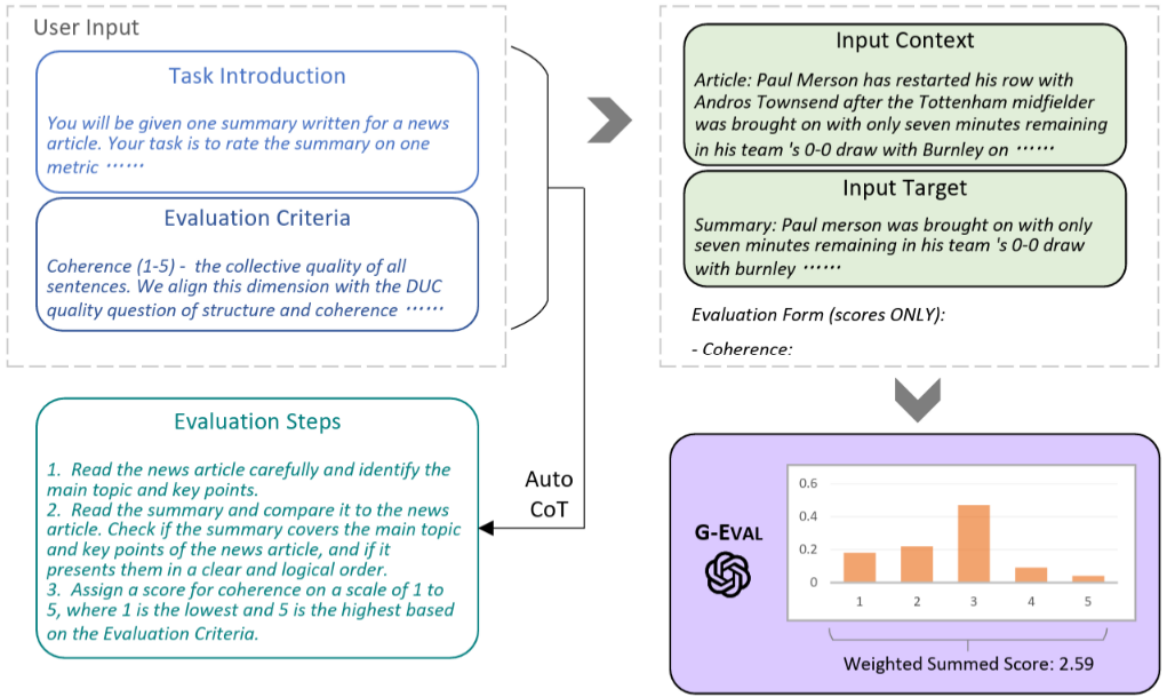
#1 GenAI as Tool to Transform Your Data

GenAI can change what we are able to analyze.

How to use?

Reminder: If we have a property of interest which is difficult to measure, GenAI can help measure it.

A closer look to LLM-as-a-judge



G-Eval: NLG Evaluation using Gpt-4 with Better Human Alignment
Liu et al. 2023

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#1 GenAI as Tool to Transform Your Data

How to reflect?

GenAI-derived data is not ground truth !

Reflecting on its reliability becomes part of your assignment.

> Examine the derived data critically: For instance, evaluate a small sample. You can use dedicated tools for this purpose like:
>
> Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences
>
> Shankar et al. 2024

> Try different prompting strategies and/or models and reflect on their reliability.

Reflect on how this expanded your findings beyond the human-only pair, the verification effort it required, and whether the value of this expansion justified the verification effort.

#2 GenAI as Collaborator to Design and Implement Your Visualization

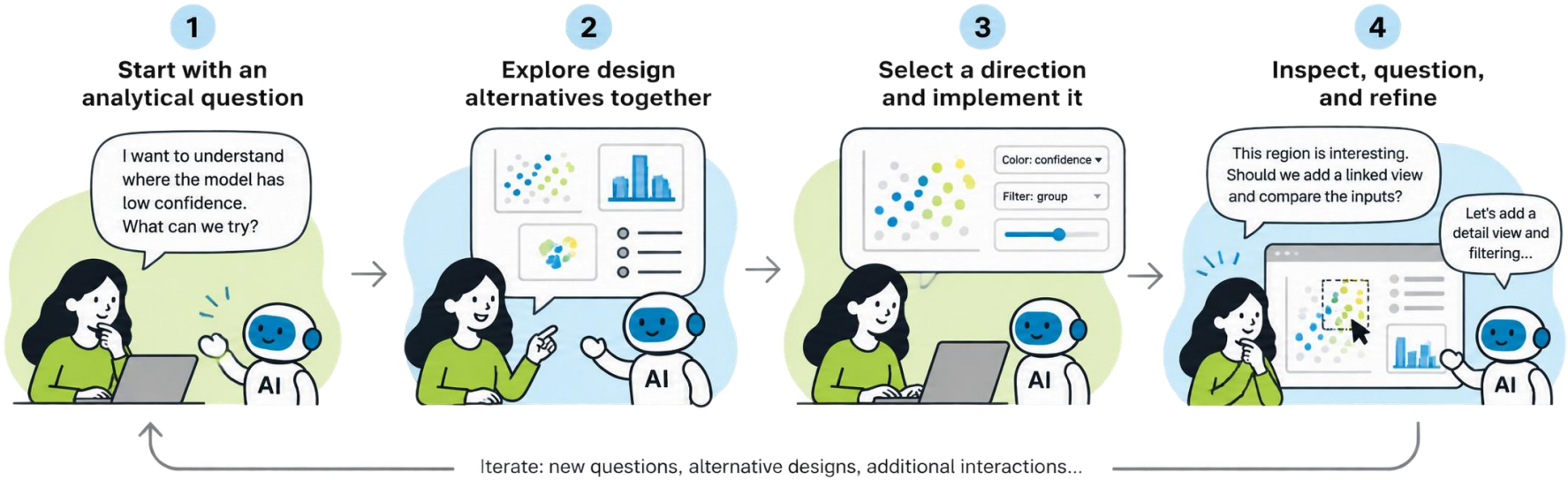
How to use?

Do not use it as a tool !

Create an interactive visualization to investigate the input-output relationships in this data ❌

Visualization design and implementation is an exploratory, progressive, and iterative process

Therefore, use it as a collaborator ✅



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#2 GenAI as Collaborator to Design and Implement Your Visualization

How to reflect?

Reflect on the collaboration itself

- Did GenAI help you explore alternatives you otherwise wouldn't have considered?
- Which design ideas came from you, and which were suggested by GenAI?
- Which suggestions were useful, rejected, or required refinement? Why?
- Compare with the human-only pair: How did the collaboration change your approaches?
- ...

#3 GenAI as Collaborator to Critique Your Design and Findings

How to use?

Invite GenAI to critically examine your design choices, and interpretations.

How to reflect?

- Did this process help uncover weaknesses, overlooked findings, or alternatives?
- Which criticisms were justified, or irrelevant? Why?
- How did this affect your confidence?
- …

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#4 GenAI as Collaborator to Steer Your Investigation

Let GenAI help you decide what to investigate next.

Share your goal, what you have explored so far, and current findings.

How to use?

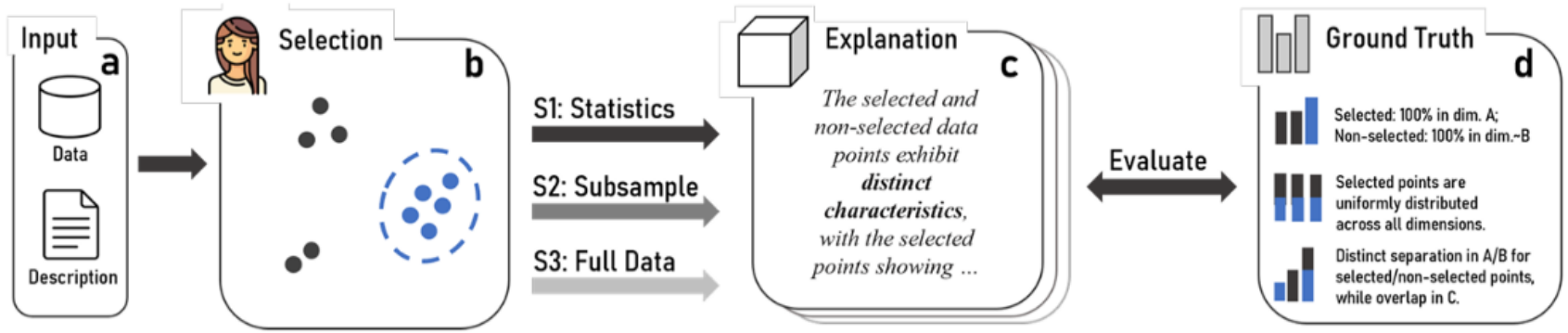
How to reflect?

- Did GenAI open up valuable directions you otherwise wouldn't have considered?
- Which suggestions were useful, or rejected? Why?
- How did this change the breadth and depth of your investigation compared with the human-only pair?
- ...

#5 GenAI as Tool to Interpret Your Observations

How to use?

GenAI can help you make sense of complex patterns by describing and contrasting the underlying data items.



LangLasso: Interactive Cluster Descriptions through LLM Explanation

Buchmuller et al. VISxGenAI IEEE VIS 2025

Further ideas:

Describing Differences between Text Distributions with Natural Language

Zhong et al. 2022

VibeCheck: Discover and Quantify Qualitative Differences in Large Language Models

Dunlap et al. 2025

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#5 GenAI as Tool to Interpret Your Observations

How to reflect?

Just like #1, GenAI interpretations is not trustworthy !

Reflecting on their reliability becomes part of your assignment.

- Reflect on how this affected your confidence in your findings.
- Compared with the human-only pair, did GenAI lead to misinterpretations or help uncover findings that the human-only pair overlooked?
- ...

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Next Week Feedback Sessions

Thu 05.11.26	12:00 - 13:30	S3 055
Thu 05.11.26	13:45 - 14:30	S3 055
Thu 05.11.26	14:30 - 15:15	Online
Fri 06.11.26	08:30 - 10:00	Online

Link to book a 10-minute slot will be posted next week on Moodle.

A team (both pairs) can book only one slot across the three sessions.

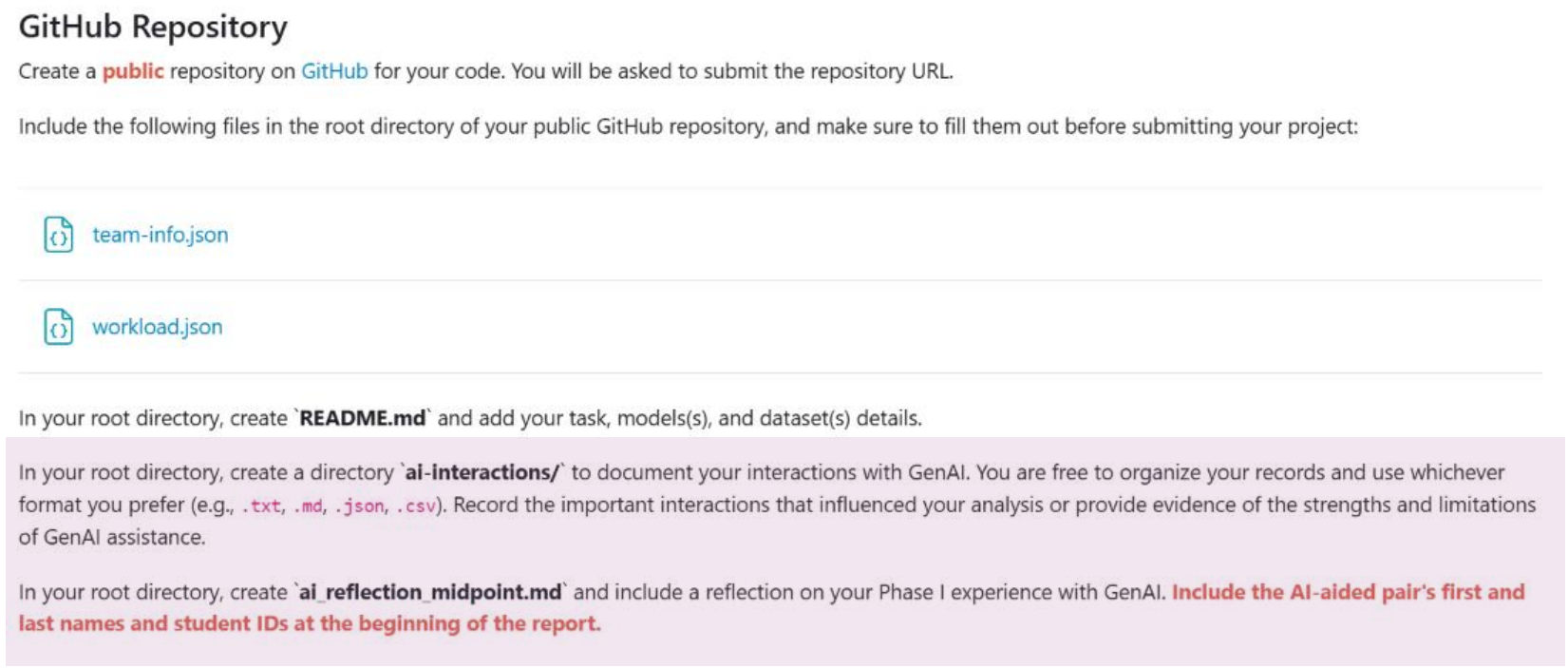
Midpoint Submission

Wed 11.11.26 23:59 Via Moodle

You will be asked to submit:

- Link to progress-update video of up to 5 minutes. **Exceeding the time limit results in a point deduction.**
  Briefly introduce the problem, summarize what each pair did and found, show key visualizations in slides, a notebook, or the UI.

- Link to your public GitHub repository.



GitHub Repository

Create a **public** repository on GitHub for your code. You will be asked to submit the repository URL.

Include the following files in the root directory of your public GitHub repository, and make sure to fill them out before submitting your project:

team-info.json

workload.json

In your root directory, create `**README.md**` and add your task, models(s), and dataset(s) details.

In your root directory, create a directory `**ai-interactions/**` to document your interactions with GenAI. You are free to organize your records and use whichever format you prefer (e.g., `.txt`, `.md`, `.json`, `.csv`). Record the important interactions that influenced your analysis or provide evidence of the strengths and limitations of GenAI assistance.

In your root directory, create `**ai_reflection_midpoint.md**` and include a reflection on your Phase I experience with GenAI. **Include the AI-aided pair's first and last names and student IDs at the beginning of the report.**

Next Session

Thu 12.11.26 13:45 - 15:15 S3 055 + online

- Practical examples on VIS4AI Phase II
- Organizational details on: Final Submission & Presentations

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Thank you